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CO₂ Valorization by Converting Flue Gas into Graphene

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Abstract

Carbon materials hold an exclusive spot among modern advanced materials with a variety of beneficial properties. However, their widespread use is hindered by challenges in scaling up production and, in some cases, a significant carbon footprint. Graphene is a 2D sheet of carbon atoms arranged in a honeycomb structure. Outstanding properties make this material of great interest for various applications, including oil extraction. The paper provides discussion of recent graphene applications and highlights its potential as a material for drilling, cementing, and stimulating fluids. The goal of this work is to summarize recent progress in developing large-scale graphene production and highlight the best approach for future improvements. Among the several graphene manufacturing strategies proposed to date, we chose one that employs plasma to convert a CO₂ and hydrogen mixture (mass ratio of 3-4 to 1) to graphene foil coated on a metal support. Such an approach surpasses other proposed technologies in terms of scalability, cost-effectiveness, and CO₂ emissions reduction. Further investigation indicated that this strategy may be applied to the mixtures of CO₂ with H₂O or H₂S rather than expensive hydrogen. The novelty of the paper lies in the development and economical assessment of the rising technology converting CO₂ to valuable material — graphene. Additionally, the possibility of using waste gas such as H₂S instead of H₂ to produce graphene was highlighted.

Introduction

Graphene is a single-atom-thick, two-dimensional (2D) sheet of carbon atoms arranged in a honeycomb crystal lattice. Graphene and graphite are both forms of carbon, but they have distinct structures and properties. Graphite consists of many layers of graphene stacked together, held by weak van der Waals forces (Figure 1). While graphite is used in applications like pencils and lubricants, graphene's unique properties make it suitable for advanced technological applications. This unique structure gives graphene its remarkable properties. Each carbon atom in graphene is covalently bonded to three other carbon atoms, forming a hexagonal pattern. This arrangement results in a planar structure that is both incredibly strong and flexible. Today this 2D sheet of carbon atoms is mainly obtained by synthetic methods from graphite or carbon nanostructures.

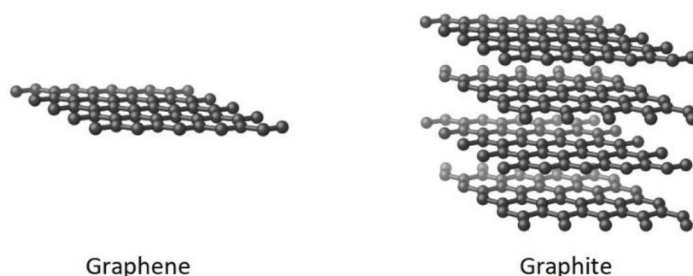


Figure 1—Graphic representation of graphene structure (single sheet of C atoms) and graphite (stacked sheets of C atoms).

Properties of Graphene

Graphene's properties are extraordinary and diverse, making it a material of great interest across various fields. Some of its key properties include:

- **Electrical Conductivity:** Graphene is an excellent conductor of electricity. Its electrons can move through the lattice with minimal resistance, making it ideal for use in electronic devices.
- **Thermal Conductivity:** Graphene conducts heat better than any other known material. This property is useful in applications requiring efficient heat dissipation.
- **Mechanical Strength:** Despite being only one atom thick, graphene is incredibly strong. It has a tensile strength of 130 GPa, which is more than 100 times stronger than steel.
- **Flexibility:** Graphene is highly flexible and can be bent, twisted, and stretched without breaking. This makes it suitable for flexible electronics and other applications where material flexibility is crucial.

Graphene Applications in the Upstream Oil Industry

Graphene's unique properties make it a valuable material for various applications in the upstream oil industry (Kremieniewski 2020). Its high strength, thermal conductivity, and chemical stability can enhance the performance of drilling fluids, cement slurries, and enhanced oil recovery (EOR) chemicals.

Drilling Fluids. Graphene can significantly improve the performance of drilling fluids (Ikram 2021). Drilling fluids, also known as drilling muds, are used to lubricate the drill bit, carry drill cuttings to the surface, and maintain wellbore stability. Adding graphene nanosheets to drilling fluids can enhance their properties in several ways (Figure 2):

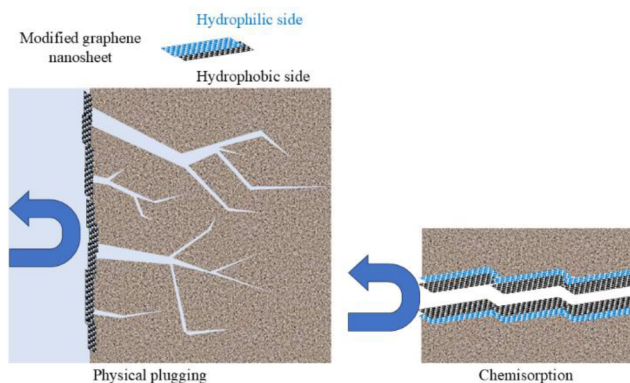


Figure 2—Proposed interaction mechanism of graphene particles with well surfaces.

- **Filtration Control:** Graphene nanosheets can plug and seal micropores in the wellbore, reducing filtration loss. This helps maintain wellbore stability and prevents the loss of drilling fluids into the formation.

- **Shale Inhibition:** Graphene's high surface area and chemical stability can inhibit shale swelling and disintegration, which are common problems in drilling operations. This property is particularly beneficial in high-temperature environments, where conventional inhibitors may fail.
- **Rheological Properties:** Graphene can improve the rheological properties of drilling fluids, making them more effective in carrying drill cuttings to the surface. This results in more efficient drilling operations and reduced downtime.

Cement Slurries. Cement slurries are used to cement the casing in oil wells, providing structural integrity and isolating different zones within the well (Haruna 2019). Adding graphene oxide to cement slurries can enhance their mechanical properties and performance:

- **Mechanical Strength:** Graphene oxide can significantly increase the mechanical strength of cement composites. Even small quantities (0.01-0.03 %) of graphene oxide, combined with a latex admixture, can result in a 100% increase in shear stress, leading to stronger and more durable cement.
- **Rheological Properties:** Graphene oxide can improve the rheological properties of cement slurries, making them easier to pump and place. This results in more efficient cementing operations and better zonal isolation.

Enhanced Oil Recovery (EOR). Enhanced oil recovery (EOR) techniques are used to increase the amount of oil that can be extracted from a reservoir. Graphene can improve the performance of EOR chemicals (Yi 2015, Haruna 2019), such as polyacrylamide, which is commonly used in polymer flooding. Following applications are possible:

- **Viscosity Enhancement:** Graphene nanosheets can act as "cross-links" for polymer molecules, helping to maintain the polymer structure under operational conditions. This can significantly enhance the viscosity of the polymer, improving its ability to displace oil from the reservoir.
- **High-Temperature Stability:** Graphene can improve the high-temperature stability of EOR chemicals. For example, during thermal aging at 80 °C for 30 days, the viscosity of the graphene-polyacrylamide composite remained unchanged, whereas the viscosity of the original polyacrylamide decreased by 59%. This makes graphene-enhanced EOR chemicals more effective in high-temperature reservoirs.

Current approaches for graphene production

Graphene synthesis methods can be broadly classified into two groups: top-down and bottom-up approaches. The top-down approach involves breaking down a precursor material to produce graphene sheets, while the bottom-up approach involves constructing graphene layers from carbon precursors. Each method has its own advantages and limitations, which are summarized in Tables 1 and 2. Understanding these methods and their drawbacks is crucial for developing more efficient and cost-effective production techniques. Detailed discussion of each method is provided below.

Table 1—Comparison of top-down approaches for graphene synthesis.

Method	Pros	Cons
Mechanical Exfoliation	High quality of graphene	High cost, hard to upscale, time-consuming
Unzipping Nanotubes	High quality of graphene	High cost, need for reagents, hard to upscale
Liquid Phase Exfoliation	Low cost, easy to operate, easy to upscale	Average graphene quality, defects
Ablation Methods	Easy to operate, can be upscaled	Average graphene quality, quality control
Arc Discharge	Good quality of graphene	Energy-intensive, hard to upscale, hard to operate
Oxidation-Reduction	Easy to upscale	Low quality of graphene, safety risks

Table 2—Comparison of bottom-up approaches for graphene synthesis.

Method	Pros	Cons
Chemical Vapor Deposition (CVD)	High quality of graphene, easy to upscale	High cost, hard to operate
Chemical Synthesis	High quality of graphene, easy to operate	High cost, hard to upscale
Laser-Related Synthesis	High quality of graphene, easy to operate, scalable	High cost

Mechanical Exfoliation

Mechanical exfoliation, also known as the "Scotch tape method," involves peeling off layers of graphene from graphite using adhesive tape. This is a straightforward method using either normal force, e.g. adhesives (Yi 2015), microscopic diamond wedges (Jayasena 2011), or shear force, e.g. ball milling or fluid dynamic methods (Chen 2012). This method can produce high-quality graphene sheets or flakes, especially when high-quality graphite is used. However, it has several drawbacks:

- **Cost:** The use of advanced and expensive machinery, high consumption of adhesives, and significant energy requirements make this method costly.
- **Time-Consuming:** Processes like ball milling can take 24 to 48 hours to produce graphene, making it inefficient for large-scale production.
- **Scalability:** Mechanical exfoliation is difficult to scale up for industrial applications due to its labor-intensive nature.

Unzipping Nanotubes

Unzipping carbon nanotubes (CNTs) is another method to produce high-quality few- or single-layered graphene. Various unzipping techniques include chemical, plasma, metal intercalation, cutting with nanoparticles or electron beam (Kumar 2021, Kim 2010, Jiat 2019). While this method yields high-quality graphene, it also has significant drawbacks:

- **Cost:** The process requires expensive and/or hazardous chemicals and equipment.
- **Precursor Production:** CNTs themselves need to be produced first, adding an additional step and cost to the process.

Liquid Phase Exfoliation

Liquid phase exfoliation involves dispersing graphite in a suitable aqueous or non-aqueous solvent, followed by exfoliation and purification. Exfoliation is commonly carried out by sonication, high-shear mixing, or high-pressure treatment (Kumar 2021). This method is considered viable for mass production due to its simplicity and low cost. However, it has its own set of challenges (Kim 2010, Jiat 2019):

- Defects: The exfoliation process often introduces a considerable number of defects in the graphene sheets, reducing their quality.
- Purification: The resulting graphene requires extensive purification to remove residual solvents and impurities.

Ablation Methods

Ablation methods, such as employing pulse laser irradiation of carbon precursors, have shown promise for large-scale industrial production of graphene. However, these methods face challenges (Kiran 2017, Jalili 2019):

- Layer Control: There is little control over the number of graphene layers produced.
- Scalability: Issues with scalability for layer ablation remain a significant hurdle.

Arc Discharge

Arc discharge is a conventional technology used for the preparation of fullerenes and CNTs and has been adapted for graphene synthesis. This method uses high-temperature plasma generated by electrical current to sublime the carbon precursor and produce stacked 2 to 10 graphene sheets. Despite its adaptability, arc discharge has several limitations:

- Energy-Intensive: The process is highly energy-intensive.
- Control: Precise control of reaction conditions is required.
- Scalability: Efforts to increase its cost-effectiveness and scalability are still ongoing.

Oxidation-Reduction

The oxidation-reduction method involves treating graphite with strong oxidants (e.g., potassium chlorate, sulfuric acid, nitric acid) to weaken the bonds between carbon sheets. The oxidized graphite is then dried in a vacuum, resulting in exfoliation and yielding graphene oxide. This method is fast and easy to upscale but has significant drawbacks:

- Toxic Reagents: The use of toxic reagents and dangerous conditions complicates the operation and poses high safety risks.
- Quality: The resulting graphene oxide contains many defects, severely damaging its quality. Further labor-intensive reduction steps are required to obtain graphene.

Chemical Vapor Deposition (CVD)

Chemical vapor deposition (CVD) is a widely used method for producing large-area graphene films. In this process, a carbon-containing gas, such as methane, is decomposed at high temperatures (650-1000°C) on a substrate, typically copper or nickel. The carbon atoms then form a graphene layer on the substrate. CVD can produce high-quality graphene suitable for industrial applications. However, it has several disadvantages:

- High Production Cost: The process is expensive due to the high temperatures and specialized equipment required.
- Purification: Further purification is needed to remove residual catalysts.
- Transfer: Transferring graphene to other substrates can be challenging and adds to the cost.

Chemical Synthesis

Chemical or organic synthesis of graphene involves assembling atomic or molecular building blocks, such as aromatic molecules. This method can produce small graphene molecules with atomically accurate and uniform nanostructures, suitable for precise and reproducible components in optoelectronic, spintronic, and other applications. Similar approach which can be exemplified is chemical reduction. Chemical reduction involves reducing graphene oxide to obtain graphene. Graphene oxide is produced by oxidizing graphite, which introduces oxygen-containing functional groups. These groups are then removed through chemical reduction, resulting in graphene. However, it is significantly expensive:

- **Cost:** The multistep organic synthesis process is costly and not suitable for large-scale production.
- **Scalability:** This method is hard to upscale for industrial applications.
- **Quality:** chemical reduction produces graphene with defects in its structure which results in lower quality compared to other methods.

Laser-Related Synthesis

Laser-related synthesis is a fast-growing method for producing graphene, particularly for specialized electronics applications. It can be subdivided into two categories: laser-induced graphene (LIG) and laser-reduced graphene (LRG). Similar to the CVD process, LIG uses a laser to irradiate a carbon-based precursor (e.g., polymers) on a substrate. This method can be performed under ambient conditions and scaled up for large-scale production. LRG is performed without a metal substrate by applying a laser beam directly onto a carbon precursor, such as graphite oxide sheets. This method yields flexible, defective, and porous graphene films, useful for applications in catalysis, adsorption, and supercapacitors. Main challenges for these methods are:

- **Cost:** Today the implementation of lasers is accompanied with high energy consumption and expensive equipment.
- **Quality:** Despite the use of complex equipment these methods produce defective porous graphene which is beneficial for some applications but is unacceptable for other.

New solutions to produce graphene

Despite the rapid developments in graphene synthesis methods, a cost-effective approach for mass production of high-quality graphene remains elusive. Each method has its own set of advantages and limitations, making it challenging to find a one-size-fits-all solution. The high cost, scalability issues, and quality control remain significant barriers to the widespread industrial implementation of graphene. Innovative production techniques are being developed to address the challenges and limitations of conventional graphene production methods. These new solutions aim to make graphene production more cost-effective, scalable, and reliable. Reducing the cost of graphene production is crucial for its commercialization. Innovative techniques, such as electrochemical exfoliation and liquid-phase exfoliation, offer cost-effective alternatives to conventional methods. These techniques can produce large quantities of graphene at a lower cost, making it more accessible for industrial applications. In the realm of advanced materials and nanotechnology, the development of new solutions is crucial for addressing global challenges such as sustainability and carbon emissions. But some graphene-producing technologies currently have a significant carbon footprint, which is detrimental to global efforts to reduce carbon emissions. For example, conventional CVD processes can have a massive carbon footprint, with estimates suggesting over 600 tons of CO₂ emissions per ton of nanocarbon produced. This high carbon footprint is a significant barrier to the widespread adoption of these technologies in a sustainable future. Therefore, there is a growing interest in developing methods to convert CO₂ into valuable materials like graphene, which can help reduce

greenhouse gas emissions while producing high-tech carbon materials. Scalability is a key factor in the commercialization of graphene. Innovative production techniques, such as liquid-phase exfoliation and plasma-assisted growth, offer scalable solutions for producing large quantities of graphene. These methods can be scaled up to meet the demands of industrial applications, making graphene more commercially viable. Up to date the main analytical technique to rate the quality of graphene is Raman spectroscopy. The intensity of distinctive Raman peaks attributed to the tangential stretching and lattice distortion can provide information on graphene quality. To ensure the consistent quality of graphene, one should prevent excess mechanical forces during synthesis, exclude impurities in reaction chamber including oxygen which could lead to oxidation of graphene.

Several innovative techniques are being explored to produce graphene more efficiently:

- **Electrochemical Exfoliation:** This method involves using an electric current to exfoliate graphene from graphite. It is a scalable and cost-effective method that can produce high-quality graphene with minimal defects.
- **Liquid-Phase Exfoliation:** In this method, graphite is exfoliated in a liquid medium using ultrasonic waves. This technique is scalable and can produce large quantities of graphene, but the quality of the graphene produced can vary.
- **Plasma-Assisted Growth:** Plasma-assisted growth involves using plasma to enhance the growth of graphene on a substrate. This method can produce high-quality graphene with fewer defects, but it requires precise control of the plasma conditions.
- **Electrolysis of molten carbonates (Figure 3):** In this method, the electrolysis of molten lithium carbonate is conducted to convert greenhouse gases into graphene. By this approach, graphene with a thickness of 1-5 carbon layers but a relatively small lateral dimension of 2-5 micrometers. Future optimization is aimed for reduction of electricity consumption and increasing the overall effectiveness of this technology (Liu 2020).
- **CO₂ to Graphene conversion:** One promising approach is the conversion of CO₂ to graphene. This method not only helps in reducing greenhouse gas emissions but also transitions the production process from using non-renewable feedstocks. A novel atmospheric pressure chemical vapor deposition (APCVD) process has been proposed (Molina-Jirón 2019), where a mixture of CO₂, argon, and hydrogen (in a 3.5:8:5 volume ratio) is used to grow graphene layers on a copper-palladium film at temperatures as high as 1000°C. Although the concept has been proven, further development is needed to improve graphene quality and reduce costs.
- **Plasma Catalysis:** Although reports on the synthesis of graphene using plasma are scarce, the combination of plasma with catalysts shows promise for nanomaterial synthesis (Centi 2021). For example, it has been demonstrated that plasma can be used in a technique similar to CVD for the controlled growth of high-quality graphene sheets on Ni or SiO₂ substrates. The benefits of using plasma include amenability for scaled processing, substantially reduced substrate and gas temperatures compared to conventional thermal CVD, and the possibility of producing good-quality graphene films (Bogaerts 2020).
- **Plasma-Enhanced Chemical Vapor Deposition (PECVD):** This method is expected to develop into a low-cost, scalable, and controllable graphene synthesis technology. An example of the beneficial application of plasma is the PECVD process (Figure 4), which has been tested for CO₂-to-graphene conversion. It has been demonstrated that good quality graphene can be produced on a copper foil from a CO₂ and hydrogen mixture (Seekaew 2022).

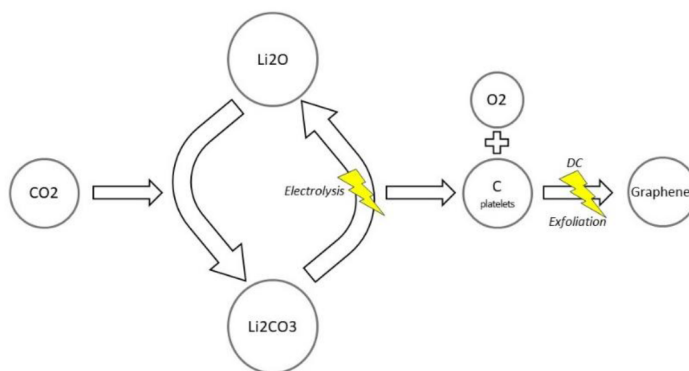


Figure 3—Process flowchart of CO₂ to graphene via lithium carbonate electrolysis.

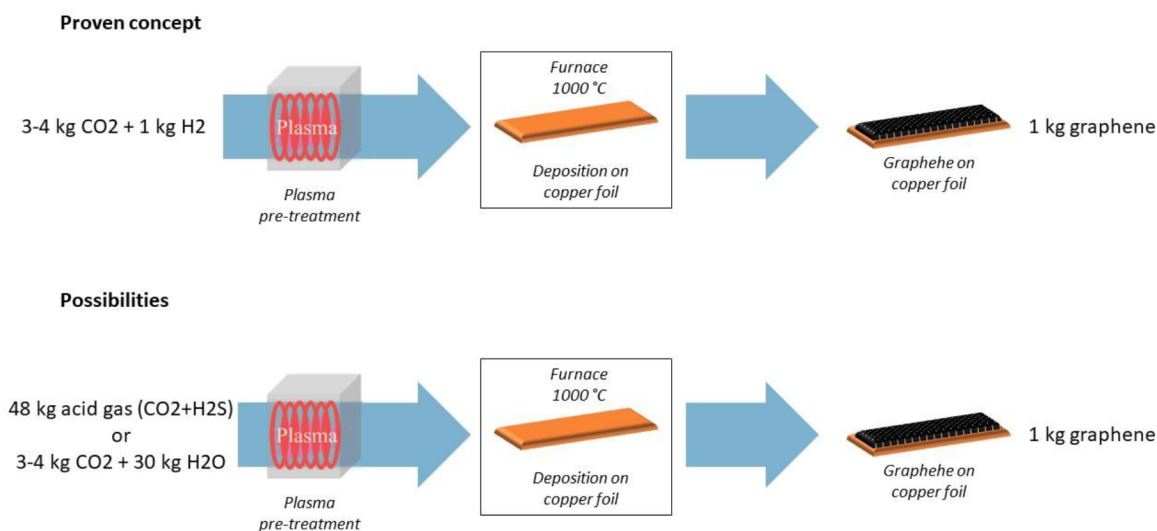


Figure 4—CVD with plasma pre-treatment for graphene production. Proven concept and future possibilities.

Conclusions

To date, numerous methods for graphene synthesis have been proposed, falling into two main categories: top-down and bottom-up approaches. Each method has its own set of drawbacks that limit its potential for large-scale implementation. As discussed, combining the Chemical Vapor Deposition (CVD) approach with plasma treatment shows significant promise for developing a low-cost, scalable, and controllable graphene synthesis technology. This concept has been proven for graphene synthesis from a CO₂ and hydrogen mixture on a lab scale. Estimations suggest that this technology could potentially be adapted for the conversion of acid gases (a mixture of H₂S and CO₂) or cheap feedstocks (water vapor and CO₂). Therefore, it is reasonable to focus laboratory research on developing plasma-enhanced chemical vapor deposition (PECVD) technologies for graphene production from CO₂.

References

- Bogaerts A., Centi G. 2020. Plasma Technology for CO₂ Conversion: A Personal Perspective on Prospects and Gap. *Frontiers in Energy Research* **8** 111
- Centi G., Perathoner S., Papanikolaou G. 2021. Plasma assisted CO₂ splitting to carbon and oxygen: A concept review analysi. *Journal of CO2 Utilization* **54** 101775
- Chen J., Duan M., Chen G. 2012. Continuous mechanical exfoliation of graphene sheets via three-roll mill. *Journal of Materials Chemistry* **22** (37) 19625–19628
- Haruna M.A., Pervaiz S., Hu Z., Nourafkan E., Wen D. 2019. Improved rheology and high-temperature stability of hydrolyzed polyacrylamide using graphene oxide nanoshee. *Applied Polymer Science* **136** (22) 47582

- Ikram R., Badrul M. J., Akhmal S., Kenanakis G. 2021. Utilization of Eco-Friendly Waste Generated Nanomaterials in Water-Based Drilling Fluids; State of the Art Review. *Materials* **14** (15) 4171
- Jalili M., Ghanbari H., Bellah S.M., Malekfar R. 2019. High-quality liquid phase-pulsed laser ablation graphene synthesis by flexible graphite exfoliation. *Journal of Materials Science & Technology* **35** (3) 292–299
- Jayasena B., Subbiah S. 2011. A novel mechanical cleavage method for synthesizing few-layer graphenes. *Nanoscale Research Letters* **6** 95
- Jiat L.X., Zhang H.B.Y., Chiew L.K., Yee L.L., Suyin G., Thangalazhy-Gopakumar S., Rigby S. 2019. Review on graphene and its derivatives: Synthesis methods and potential industrial implementation. *Journal of the Taiwan Institute of Chemical Engineers* **98** 163–180
- Kim K., Sussman A., Zettl A. 2010. Graphene nanoribbons obtained by electrically unwrapping carbon nanotube. *ACS Nano* **4** (3) 1362–1366
- Kiran G. R., Chandu B., Acharyya S.G., Rao S.V., Vadali V., Srikanth S. S. 2017. One-step synthesis of bulk quantities of graphene from graphite by femtosecond laser ablation under ambient condition. *Philosophical Magazine Letter* **97** (6) 229–234
- Kremieniewski M. 2020. Influence of Graphene Oxide on Rheological Parameters of Cement Slurries. *Energies* **13** (20) 5441
- Kumar N., Salehiyan R., Chauke V., Botlhoko O. J., Setshedi K., Scriba M., Masukume M., Ray S.S. 2021. Top-down synthesis of graphene: A comprehensive review. *FlatChem* **27** 100224
- Liu X., Wang X., Licht G., Licht S. 2020. Transformation of the greenhouse gas carbon dioxide to graphene. *Journal of CO₂ Utilization* **36** 288–294
- Molina-Jirón C., Chellali M.R., Kumar C.N., Kübel C., Velasco L., Hahn H., Moreno-Pineda E., Ruben M. 2019. Direct Conversion of CO₂ to Multi-Layer Graphene using Cu–Pd Alloy. *ChemSusChem* **12** (15) 3509–3514
- Seekaew Y., Tammanoon N., Tuantranont A., Lomas T., Wisitsoraat A., Wongchoosuk C. 2022. Conversion of Carbon Dioxide into Chemical Vapor Deposited Graphene with Controllable Number of Layers via Hydrogen Plasma Pre-Treatment. *Membranes (Basel)* **12** (8) 796
- Yi C., Shi G. 2014. Functional Gels Based on Chemically Modified Graphene. *Advanced Materials* **26** (24) 3992–4012
- Yi M., Shen Z. 2015. A review on mechanical exfoliation for the scalable production of graphene. *Journal of Materials Chemistry A* **3** (22) 11700–11715